

Ethiopian Birr Banknote Denomination Recognition

Capstone Project Summary

Using Deep Learning and Transfer Learning

September 2026

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1. Problem Definition

Currency recognition is a fundamental task in financial transactions, retail automation, and accessibility tools for visually impaired individuals. In Ethiopia, banknotes of different denominations share similar physical dimensions and color characteristics, making manual identification challenging - particularly for people with visual impairments.

Currently, there is a lack of automated, accessible tools specifically designed to recognize Ethiopian Birr banknote denominations. Existing currency recognition systems are primarily built for widely circulated currencies such as USD, EUR, and INR, leaving a significant gap for Ethiopian currency users.

This project addresses the need for an intelligent system that can automatically identify Ethiopian Birr banknote denominations (5, 10, 50, 100, and 200 Birr) from images, while also rejecting non-Birr images as "Other". The system uses deep learning - specifically Convolutional Neural Networks (CNNs) and transfer learning - to classify banknote images captured via camera or uploaded by the user.

The solution is deployed as a web application with voice output, making it practical and accessible for real-world use, including assistive technology for visually impaired users.

2. Objectives

General Objective

To develop a deep learning-based system that accurately recognizes and classifies Ethiopian Birr banknote denominations from images while rejecting non-Birr images, and to deploy it as an accessible web application.

Specific Objectives

- Collect and preprocess a dataset of Ethiopian Birr banknote images covering five denominations.
- Incorporate a diverse "Other" class using natural images to enable the model to reject non-Birr inputs.
- Build and train a baseline CNN model for multi-class image classification.
- Apply transfer learning using MobileNetV2 to improve classification accuracy.
- Compare the performance of the custom CNN versus the transfer learning approach.
- Deploy the trained model as an interactive web application using Streamlit.
- Implement voice output (text-to-speech) for accessibility.

3. Dataset Description

The project uses two datasets combined into a 6-class classification problem: five Ethiopian Birr denomination classes plus an "Other" class for non-Birr images.

3.1 Ethiopian Currency Dataset

The primary dataset consists of 1,919 images of Ethiopian Birr banknotes, organized into 11 sub-classes (front and back of each denomination, plus a background class). The images were collected under varying lighting conditions and angles to improve model robustness.

For model training, front and back images of each denomination are mapped to the same class label, resulting in 5 denomination classes.

Denomination	Train (F)	Train (B)	Test (F)	Test (B)	Total
5 Birr	148	148	36	36	368
10 Birr	148	148	36	36	368
50 Birr	148	148	36	36	368
100 Birr	148	148	36	36	368
200 Birr	148	148	36	36	368
Background	64	-	15	-	79

3.2 Natural Images Dataset ("Other" Class)

To enable the model to reject non-Birr images, the Natural Images dataset (6,899 images across 8 everyday object categories) is used as the "Other" class. This ensures the model does not force a denomination prediction on arbitrary images.

Category	Images
Airplane	727
Car	968
Cat	885
Dog	702
Flower	843
Fruit	1,000
Motorbike	788
Person	986
Total	6,899

3.3 Combined Dataset Summary

Split	Birr Images	Other Images	Total
Training	1,480	5,519	6,999
Testing	360	1,380	1,740
Total	1,840	6,899	8,739

Data Preprocessing

- Images resized to 224 x 224 pixels to match the MobileNetV2 input requirements.
- Pixel values normalized from [0, 255] to [0, 1] range.
- 80/20 train-validation split applied to the training set.

Data Augmentation

To reduce overfitting and increase data diversity, the following augmentation techniques were applied during training:

- Random rotation (up to 5 degrees)
- Random zoom (up to 10%)
- Random horizontal and vertical translation (up to 10%)

4. Methodology

The project follows a systematic deep learning workflow consisting of the following phases:

Phase 1: Data Collection and Exploration

Images of Ethiopian Birr banknotes were collected and organized into labeled directories. The Natural Images dataset was added as a diverse "Other" class. Exploratory Data Analysis (EDA) was performed to verify class distributions, visualize sample images, and ensure data quality.

Phase 2: Preprocessing

All images were resized to 224x224 pixels and normalized. Labels were encoded as integers (0-5) for sparse categorical cross-entropy loss: 0=5 Birr, 1=10 Birr, 2=50 Birr, 3=100 Birr, 4=200 Birr, 5=Other. A validation set was created from the training data to monitor overfitting.

Phase 3: Model Development

Two models were developed and compared:

- A custom CNN built from scratch (baseline model).
- A MobileNetV2-based transfer learning model (final model).

Both models were trained with data augmentation and evaluated on the held-out test set.

Phase 4: Evaluation

Models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices to assess overall and per-class performance across all 6 classes.

Phase 5: Deployment

The best-performing model (MobileNetV2) was saved and integrated into a Streamlit web application with camera input, image upload, and voice output capabilities. The app handles "Other" classifications gracefully with appropriate user messaging.

5. Model Architecture

5.1 Baseline CNN

The baseline model is a custom Sequential CNN with four convolutional blocks, followed by a Global Average Pooling layer and a dense classifier. Each convolutional block consists of:

- Conv2D layer with increasing filters (32, 64, 128, 256)
- Batch Normalization
- ReLU Activation
- MaxPooling2D (2x2)

The classification head includes a 128-unit Dense layer with ReLU, a Dropout layer (0.3), and a 6-unit Dense layer with Softmax activation.

5.2 MobileNetV2 Transfer Learning (Final Model)

The final model leverages MobileNetV2, a lightweight and efficient CNN pre-trained on the ImageNet dataset (1.4 million images, 1,000 classes). The architecture consists of:

- Data augmentation layer (rotation, zoom, translation)
- Rescaling layer to convert pixel values to [-1, 1] range (MobileNetV2 requirement)
- Frozen MobileNetV2 base (feature extractor, not trainable)
- Global Average Pooling 2D
- Dense layer (128 units, ReLU activation)
- Dropout (0.3)
- Output Dense layer (6 units, Softmax activation)

Training Configuration

Parameter	Value
Optimizer	Adam
Learning Rate	0.001 (with ReduceLROnPlateau)
Loss Function	Sparse Categorical Cross-Entropy
Batch Size	32
Epochs	20 (with EarlyStopping, patience=5)
Input Size	224 x 224 x 3
Validation Split	20% of training data
Number of Classes	6

6. Training and Evaluation

6.1 Baseline CNN Training

The baseline CNN was trained for 20 epochs on the 6-class dataset (5 Birr denominations + Other). While training accuracy improved steadily, the model showed significant overfitting with lower validation accuracy. Nevertheless, with the addition of the "Other" class, the baseline achieved 95% test accuracy - a substantial improvement over the previous 5-class version (64%).

6.2 MobileNetV2 Training

The MobileNetV2 transfer learning model converged much faster and showed strong generalization. Key observations:

- Validation accuracy reached ~94% by epoch 7.
- ReduceLROnPlateau reduced the learning rate when validation loss plateaued.
- EarlyStopping restored the best weights to prevent overfitting.
- The gap between training and validation accuracy remained small, indicating good generalization.
- The "Other" class was classified with near-perfect accuracy (F1 = 1.00).

7. Results

7.1 Model Comparison

Metric	Baseline CNN	MobileNetV2
Test Accuracy	95%	99%
Test Loss	0.128	0.037
Overfitting	Moderate	Minimal
Classes	6	6

7.2 MobileNetV2 Classification Report

Class	Precision	Recall	F1-Score	Support
5 Birr	0.88	0.94	0.91	72
10 Birr	1.00	0.96	0.98	72
50 Birr	0.99	0.99	0.99	72
100 Birr	0.93	0.89	0.91	72
200 Birr	0.93	0.94	0.94	72
Other	1.00	1.00	1.00	1,380
Weighted Avg	0.99	0.99	0.99	1,740

The MobileNetV2 model achieved 99% overall accuracy on the test set, with consistently high precision, recall, and F1-score across all six classes. The "Other" class achieved perfect classification (F1 = 1.00), demonstrating the model's ability to reliably reject non-Birr images. Among Birr denominations, the 50 Birr class achieved the highest performance (F1 = 0.99), while 5 Birr and 100 Birr showed slightly lower but still strong results (F1 = 0.91).

7.3 Baseline CNN Classification Report

Class	Precision	Recall	F1-Score	Support
5 Birr	0.76	0.57	0.65	72
10 Birr	0.98	0.75	0.85	72
50 Birr	0.76	0.90	0.82	72
100 Birr	0.68	0.86	0.76	72
200 Birr	0.85	0.86	0.86	72
Other	0.99	0.99	0.99	1,380

Weighted Avg	0.96	0.95	0.95	1,740
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The baseline CNN achieved 95% test accuracy. While the "Other" class was classified with high accuracy (F1 = 0.99), performance on Birr denominations was more variable, with the 5 Birr class showing the lowest F1-score (0.65).

8. Web Application

The trained MobileNetV2 model is deployed through a Streamlit web application that provides an intuitive interface for banknote recognition. The application is designed with accessibility in mind, featuring automatic voice output of prediction results.

Application Features

- Image Upload: Users can upload banknote images in JPG, JPEG, or PNG format.
- Camera Capture: Users can take photos directly from their device camera.
- Real-time Prediction: The model processes the image and displays results instantly.
- Confidence Score: Shows prediction confidence as a percentage.
- Probability Breakdown: Displays per-class probabilities with progress bars for all 6 classes.
- Voice Output: Automatically speaks the predicted denomination using the Web Speech API. For non-Birr images, it announces that the image does not appear to be an Ethiopian Birr banknote.
- Confidence Alerts: Provides feedback based on confidence level - high (>90%), moderate (70-90%), or low (<70%).
- "Other" Detection: Non-Birr images are clearly labeled as outside the supported categories, preventing false denomination predictions.

How It Works

1. The user selects an input method (upload or camera).
2. The uploaded/captured image is converted to RGB and resized to 224x224 pixels.
3. Pixel values are normalized to the [0, 1] range.
4. The image is fed through the MobileNetV2 model for prediction.
5. The predicted class (denomination or "Other"), confidence, and probability distribution are displayed, and the result is spoken aloud.

Running the Application

```
streamlit run app.py
```

9. Technologies Used

Category	Technology
Programming Language	Python 3.10+
Deep Learning	TensorFlow / Keras
Pre-trained Model	MobileNetV2 (ImageNet)
Web Framework	Streamlit
Image Processing	Pillow (PIL), NumPy
Data Analysis	Pandas, Matplotlib, Seaborn
Machine Learning	Scikit-learn
Voice Output	Web Speech API (browser-native)
Development	Jupyter Notebook
Version Control	Git / GitHub

10. Dataset Sources

The following datasets were used in this project:

1. Ethiopian Currency Dataset

Contains 1,919 images of Ethiopian Birr banknotes (5, 10, 50, 100, and 200 Birr denominations) with front and back views.

Source: <https://www.kaggle.com/datasets/iyasusaketa/ethiopian-note-currency-dataset>

2. Natural Images Dataset

Contains 6,899 images across 8 categories (airplane, car, cat, dog, flower, fruit, motorbike, person). Used as the "Other" class to train the model to reject non-Birr images.

Source: <https://www.kaggle.com/datasets/prasunroy/natural-images>

11. Conclusion and Future Work

Conclusion

This project successfully demonstrates that deep learning, particularly transfer learning with MobileNetV2, can effectively classify Ethiopian Birr banknote denominations with high accuracy (99%). The addition of an "Other" class using the Natural Images dataset significantly improved the system's practical utility by enabling it to reject non-Birr images rather than forcing incorrect denomination predictions.

The comparison between the custom CNN (95% accuracy) and the MobileNetV2 model (99% accuracy) clearly shows the advantage of leveraging pre-trained features from large-scale datasets like ImageNet.

The deployed Streamlit web application provides a practical, user-friendly interface that supports both image upload and camera capture, with voice output for accessibility. The system is lightweight and can run on standard hardware without GPU requirements.

Key Achievements

- Built and compared two deep learning models for Ethiopian Birr denomination recognition with "Other" class rejection.
- Achieved 99% test accuracy using MobileNetV2 transfer learning across 6 classes.
- Incorporated 6,899 natural images as an "Other" class for robust non-Birr image rejection.
- Deployed a fully functional web application with camera and voice capabilities.
- Created an accessible tool that can assist visually impaired users.

Limitations

- The Birr dataset size is relatively small (1,919 images), which may limit generalization to rare variants.
- The model only recognizes five denominations and does not detect counterfeit notes.
- Performance may degrade under poor lighting conditions or extreme angles.
- The voice output relies on browser-native Web Speech API, which may vary across browsers.
- The "Other" class is trained on 8 specific object categories; unseen object types may occasionally be misclassified.

Future Work

- Expand the dataset with more diverse images (different lighting, backgrounds, wear conditions, and partially occluded notes).
- Add support for the 1 Birr note and coin denominations.
- Implement counterfeit detection as an additional classification layer.
- Fine-tune the MobileNetV2 backbone (unfreeze later layers) for potentially higher accuracy on Birr denominations.
- Deploy the application as a mobile app for greater accessibility.
- Add Amharic language support for the voice output and user interface.
- Explore model optimization (quantization, TFLite) for on-device inference.
- Expand the "Other" class with more diverse image categories to further improve rejection robustness.

References

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